CHAPTER FIVE
THE CLIMATE OF ETHIOPIA AND THE HORN

- What is weather ? what is climate?

- Weather is a **daily atmospheric c**ondition

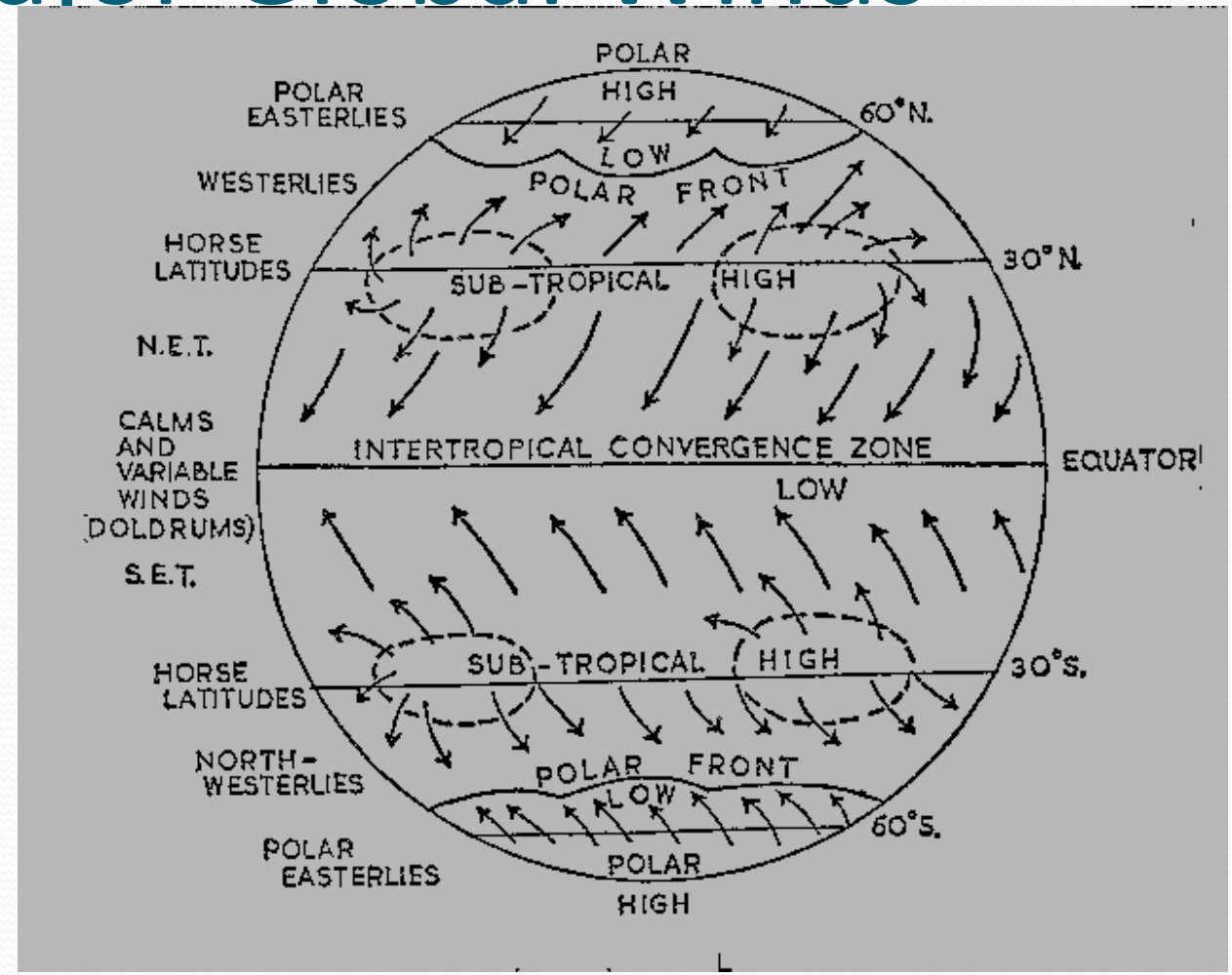
Eg. Hotness or coldness, rainy or cloudiness, sunniness, windiness, of air you are feeling **<u>on the daily base</u>**

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- ***Climate refers*** the state of the atmosphere **over long time periods**

- The climate of Ethiopia is mainly controlled by the seasonal migration of the ***ITCZ*** *and associated* ***atmospheric circulations*** *as well as by the complex* ***topography*** *of the country*.
- **ITCZ(Inter Tropical Convergence Zone)** is a low pressure zone formed by the convergence of Northeast Trade winds and the Equatorial Westerlies.
- Depending on the position of the overhead sun ITCZ shifts within the tropics.

Major Global Winds



- **Elements of Weather and Climate**
  - Temperature ,
  - Precipitation and humidity
  - Wind and air pressure

- **Rainfall** is a form of **precipitation** that comes in form of water drops.
- ***Temperature****: - Is the degree of* ***hotness or coldness*** *of the atmosphere.*
- ***Wind****: - This is moving air (air in motion). Winds blow from areas of high pressure to those of low pressure.*
- ***Atmospheric Pressure****: Is the* ***force exerted by the weight of air*** *on the earth's surface. Cold air is heavier (dense) and therefore exerts more pressure than warm air that tends to rise.*
- ***Humidity:*** *Is the amount of* ***water vapour present in the atmosphere***. Water vapour results from evaporation or transpiration.

- **Controls of Weather and Climate** are:-

  - ***Latitude*** *(distance from equator)/angle of the sun*
  - ***Altitude*** *(the height above sea level)and mountain barriers*
  - ***Distance from the sea*** *(land and water distribution)*
  - ***Ocean current :***
  - ***winds and atmospheric pressure***

- N.B. ***winds and atmospheric pressure*** *are both control and elements of climate*

- Ethiopia has varieties of climate.
- The **areal** and **seasonal** distribution of climate in Ethiopia is controlled by the following **physical factors**

1. Latitude (Distance from equator)

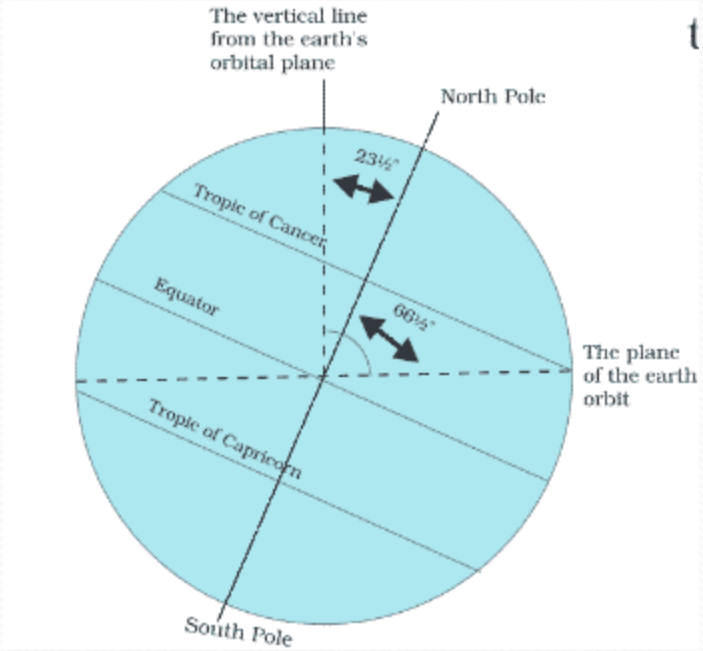
- It is related with the **angel of the sun's** rays.
- Latitude determines both the length of day through out the year and the intensity and possible duration of sunlight received.
- Generally, temperature **decreases** when we go from **Lower latitude (equator)** to polar (higher latitudes) and vice versa.
- This is due to the rays of the sun are **vertical** on equator and **slanting (oblique)** in temperate polar zones.

Effects of latitude in Ethiopia

- *Every parts of the country receive over head sun **twice** a year (I.e. first between April and June and second between June and September).*
- *In Ethiopia , there is very **high daily range** of temperature. And **low annual range** of temperature.*
- *There is very **little difference between the lengths of day and night**. Thus, the longest days occur in summer, while the shortest in winter*

Inclination of the Earth's Axis

It is formed due to the earth’s revolution around the sun
It produces a change in the **directness** of the sun's rays; which in turn causes **differences in length of day and seasons.**



Equinoxes and Solstices

- June 21 (23 ½ *in the Tropic of Cancer)*
- Sept 23 March 21
- December 22 **(Tropic of Capricorn)**

Equinoxes and Solstices

An ***equinox*** is when the length of day and night are equal (the sun at equator).

Equinox appears twice a year.

- There are two major equinoxes
- 1. ***The Vernal (spring) equinox:*** This equinox experiences in **Northern Hemisphere** when the sun is exactly above the equator.
- **March 21** marks the offset of the vernal equinox**.**

- ***2. The Autumn equinox:*** It appears to happen when the visible sun moves **south** across the celestial equator on **23rd of September**. It marks the beginning of Autumn season.

- ***Solstice is*** an event when unequal length of days and nights in the hemispheres.

Solstice is an event when unequal length of days and nights in the hemispheres.

A. ***The summer Solstice: on June 21st,*** *the sun experiencing longest daylight of the year in the northern hemisphere . It is the astronomical first day of summer in the* ***Northern Hemisphere****.*

The sun is at its highest position in the noonday sky, directly above 23 ½ in the Tropic of Cancer.

- ***B. The winter solstice:***

On December 22, *the sun experiencing longest daylight of the year in* ***Southern Hemisphere.*** It occurs when the sun is directly over the Tropic of Capricorn(23 ½ ° south of the equator*)*

2. Altitude – is by far the most important control of climate in Ethiopia

- ***In troposphere lapse rate(***temperature changes per unit) ***is 6.5°C per kilometer rise in altitude.***
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- In general, in Ethiopia as altitude increase
  - *Temperature decrease*
  - *Rainfall increase*
  - *Pressure decreases*
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Spatiotemporal Distribution of Temperature

- Every part of the country to experience **overhead sun twice a year.**
- Temperature decreases towards the interior highlands. Mean annual temperature varies from over 30 0C in the tropical lowlands to less than 10 0c at very high altitudes

- The **lowest** mean annual temperatures are recorded (Bale Mountains) and
- The highest mean maximum temperature in the country is recorded in the **Afar Depression.**

Temperature Zones of Ethiopia by altitude

Altitude (meters	Temperature zones		Mean annual temp^{0}C
	Global	Local	
3300 and above	Alpine (cool)	Wurch (kur)	below10^0
2,300-3300	Cool Temperature	Dega	10 – 15
1500 – 2300	sub-tropical (Temperature)	Woina Dega	15 – 20
500 – 1500	Tropical (warm temperate)	Qolla	20 – 25
Below 500	Desert (hot)	Bereha	>25

- Most parts of Ethiopia get its **maximum temperature** when the sun is overhead in the north of equator (**from March to June**).
- Paradoxically, **low temperature** are recorded from **November to February** (when the sun is overhead in the south of equator)
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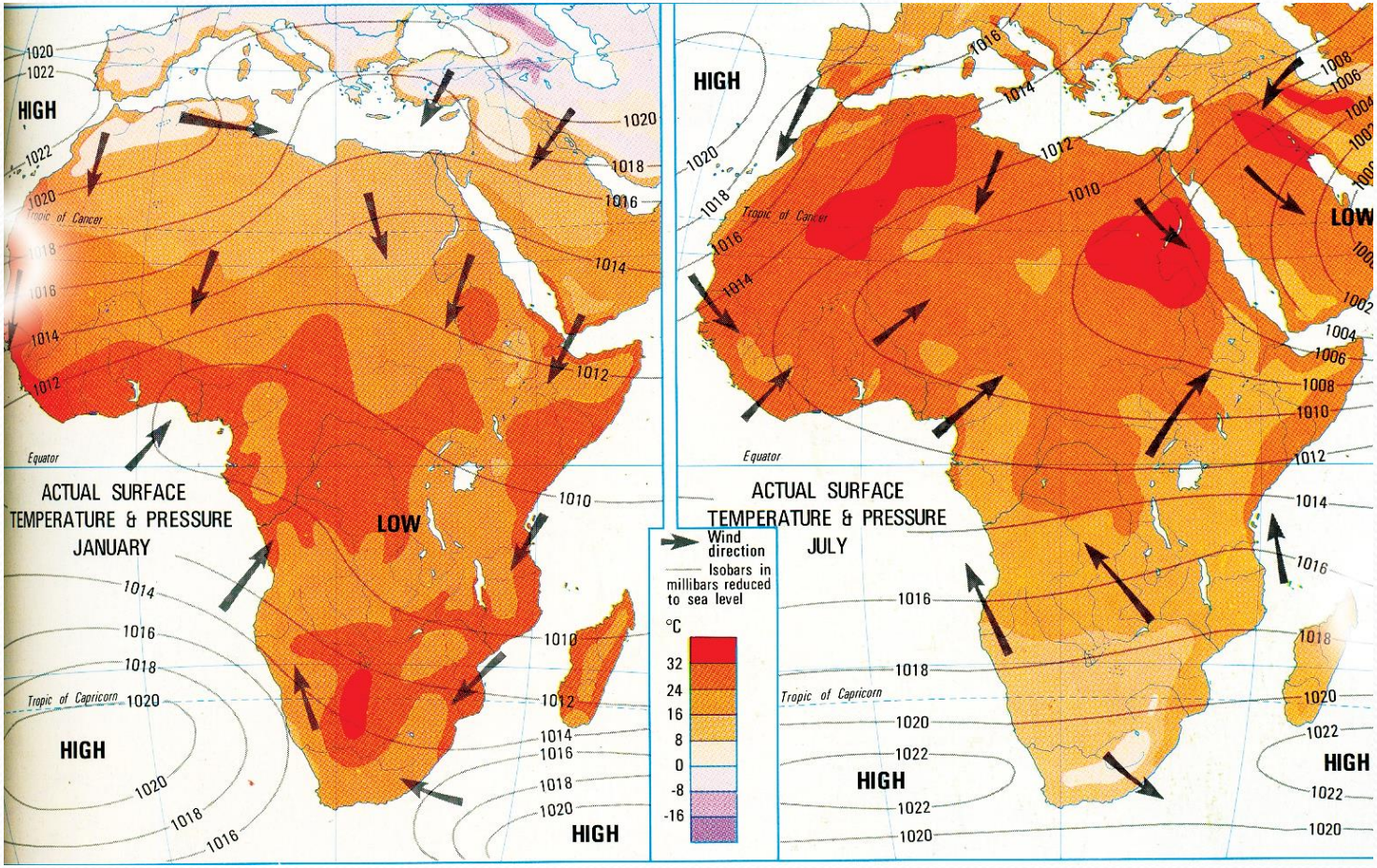
- **The *southern and southwestern highlands*** *experience reduced temperature.*
- .The direction of rain bearing winds (leeward or windward side) determines the temperature variations in mountainous regions.

Spatiotemporal Distribution of Rainfall

- It is the result of the strength and nature of **prevailing weather systems** develop from
  - *- the apparent movement of the overhead sun*
  - *- prevailing winds*
  - *- the associated inter-tropical convergence zone*
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- Depending on the position of the overhead sun, ITCZ shifts within the tropics.
- In **July**, Ethiopia comes under the influence of the **Equatorial westerlies and Easterlies winds** when overhead sun's position is at the Tropic of cancer.
- In **January**, sun's position shifts to the tropic of Capricorn leaving the region for the prevalence of the **Northeast Trade winds** (dry winds). During this time only the Eritrean coastal lands and the Afar region receive rain. In most of Ethiopia, western Eriteria and Somalia, **it becomes dry** season.

The position of the overhead sun January	The position of the overhead sun July



Seasonal or Temporal Variability of rainfall

- *What winds bring summer rainfall for Ethiopian highlands?*

Summer Season (June, July, and August) Rainfall region

- From mid-June to mid-September, Locally known as 'keremt' or wet season of Ethiopia. The overhead sun is north of equator
- ITCZ are in the south West Africa while high pressures (anticyclones) are over Atlantic and Indian oceans around tropic of Capricorn.
- the major **wind system** that bring summer rain in Ethiopia are the **Equatorial Westerlies** (Guinea monsoon winds) and **south easterlies.**
- **Atlantic** and **Indian oceans** are sources of 'keremt' rainfall but Atlantic is the main source.
- '**Keremt**' rains covers **most parts** of Ethiopia **Except south east lowlands** and **Afar depression.**
- The most rainfed regions of Ethiopia are the **south Western High lands.**
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Autumn (September, October and November)

- Tseday /Meher season
  - **South eastern Low lands** get rainfall in **Autumn**
- In autumn the ITCZ shifts towards the equator weakening the **equatorial westerlies**.
- During this season, *the south easterlies* from Indian Ocean showers the **lowlands in southeastern part** of Ethiopia.

Winter (December, January and February)

- It is Bega (dry season in Ethiopia)
- The sun is located in the southern hemisphere
- As a result the ITCZ shifts and located over southern Ethiopia.
  - **Northeast trade winds** invade Ethiopia. Northeast trade winds are dry because it rises from land masses.
  - Thus, winter is the driest season in Ethiopia

Spring (March, April and May)

- Locally known as ‘Belg’ or ‘Little rains’ season
- Most winds (on – shores) from the Indian Ocean invade the southern part of Ethiopia for the second time.
- The south east trade winds blow from the high pressure areas over the Gulf of Aden and the Indian Ocean.
- **SE trade winds** bring the main rains to the **south east lowlands and adjacent highlands.**
- thus, south **east** lowlands and adjacent highlands have two rains period: namely, autumn and spring rains:

Summary of Seasonal or Temporal Variability of rainfall

Season	Wind System *Bring The Main Rains*	Source	Parts Covers
Summer (June, July, and August)	**Equatorial Westerlies And South Easterlies.**	**Atlantic And Indian Oceans . Atlantic is** the main source.	**Most Parts of Ethiopia** Except **SE Lowland And Afar Depression**
Autumn /Tseday /Meher (Sept, Oct. and Nov)	***The South Easterlies***	**Indian Ocean**	**Southeastern Lowlands**
Winter/ bega *(December, January and February*	***Northeast Trade Winds***	**dry winds** rises from land masses	**Dry season**
Spring(March, April and May)	***SE Trade Winds***	**Indian Ocean**	***The South East Lowlands And Adjacent Highlands.***

Agro Ecological Zones of Ethiopia

Zones	Altitude (m)	Mean annual rainfall (mm)	Length of growing periods (days)	Mean annual temperature (^{0}C)	Area share (%)
Wurch (cold to moist)	>3,200	900-2,200	211–365	Below 10	0.98
Dega (cool to humid)	2,300 - 3,200	900-1,200	121–210	≥11.5–17.5	9.94
Weyna Dega (cool sub humid)	1,500 – 2300	800-1,200	91–120	>17.5 – 20.0	26.75
Kola (Warm semiarid)	500 - 1,500	200-800	46–90	>20.0 – 27.5	52.94
Berha (Hot arid)	<500	Below 200	0–45	>27.5	9.39

Climate Change/Global Warming: Causes, Consequences and Response Mechanisms

- **Climate change** refers to any change in climate over time, due to either natural variability or human activities.

- **Climate Variability** is defined as **variations** in the mean state and other statistics of the climate on all **temporal** and **spatial** scales

Current Trends of Climate in Ethiopia

- In Ethiopia, **12** extreme drought events were recorded between **1900** and **2010**. The majority of these resulted in famines.
- Among the 12, seven of the drought events occurred since 1980.

Trends in Temperature Variability

- Mean annual *temperature has shown 0.2°C to 0.28°C rise per **decade** over the last 40-50 years.*
- *A rise in average temperature of about **1.3°C** has been observed between 1960 and 2006.*
- *The rise has **spatial** and **temporal** variation.*
- *Higher rise in temperature noted in **drier areas** in northeast and southeast part of the country.*
- *Notably the variability is higher in **July-September.***

Trends in Rainfall Variability

- **Precipitation** has remained fairly stable over the last 50 years when **averaged over the country**.
- **Rainfall variability** is increasing in many parts of the country.
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Causes of Climate Change

A. Natural Causes

1. ***Earth orbital changes:*** Changes in the tilt of the earth.

2. ***Energy Budget :*** the star has been very gradually increasing its amount of radiation so that it is now 20% to 30% more intense than it was once.

3. ***Volcanic eruptions:*** releases large volumes of sulphur dioxide, C02, water vapor, dust, and ash into the atmosphere.

- The release of large volume of gases and ash can **increase planetary reflectivity** causing **atmospheric cooling**.

B. Anthropogenic Causes

- The warming of earth planet in the past 50 years is **majorly driven** by human activities.

- The **industrial activities** that our modern civilization depends upon have raised atmospheric **carbon dioxide levels**.

- Human induced greenhouse gases such as **carbon dioxide, methane and nitrous oxide** have caused much of the observed increase in Earth's temperatures over the past 50 years.
- The decomposition of wastes in landfills, synthetic compounds of manufacturing, clearing of land for agriculture, industrial activities, and other human activities have increased concentrations of greenhouse gases.

Consequences of Climate Change

1. Impacts on human health:

- The change can cause increased heat related **mortality**
- Greater frequency of **infectious disease epidemics** following floods and storms, and
- Substantial **health effects** following population displacement to escape extreme weather events.
- Climate change also raises the **incidence malaria**.

2. Impact on water resources:

i. *Climate change is leading to* ***melting of snow*** *and glaciers that* **increases** *rise in sea level,*

ii. increase ***drought*** *and floods, distorts wind flow pattern,* ***decreases water table.***

More frequent and longer droughts reduce the amount of run-off into rivers, streams and lakes.

3. Impact on Agriculture:

- **The** Changes affect agricultural production, increases physiological stress and fodder quality and availability.

4. Impact on Ecosystem: *climate change affects the success of species, population, and community adaptation.*

Climate change Response Mechanisms

- There are three response mechanisms to climate change namely ***mitigation, adaptation and resilience.***

- *1.* **Mitigation and its Strategies : reduce and control greenhouse** gas emissions

- **Some mitigation measures**
  - Practice Energy efficiency
  - Increase the use of renewable energy such as solar
  - Efficient means of transport implementation: electric public transport, bicycle, shared cars etc

2. Adaptation and Resilience and its Strategies

- **Adaptation** involves adjusting to actual or expected future climate change.
- **Resilience** refers to the rate at which an individual, community or a country **recovers from** such setbacks.

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- **Some of the major adaptation strategies include:**

- Building flood defenses,
- Installing water-permeable pavements to better deal with floods and storm water
- Improve water storage and use are some of measures taken by cities and towns.
- Landscape restoration and reforestation,
- flexible and diverse cultivation